

MTH1001 Mathematical Structures 2025–26, Term 1
Problem Sheet 2

This sheet is for discussion in the week 5 tutorials (Wednesday 22nd October). Be ready to discuss (at least) Questions A2(c)(d), A3(a)(b), B2(b)(c), B3 in the tutorial.

Your answers should use mathematical notation correctly and be written in logically correct sentences, including words such as “let”, “if”, “then”. You should introduce any variables that occur in your argument.

SECTION A

These are fairly routine questions of a similar difficulty to those on tests and the mid-term exam.

- A1. Which of the following define a function with the indicated domain and codomain? In cases which do *not* define a function, explain why not.

(a) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{x^2 + 3}$;

(b) $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = \frac{1}{x^2 + 3x + 2}$;

(c) $h : \mathbb{Z} \rightarrow \mathbb{Z}$, $h(n) = \frac{1}{3}n(n + 1)$;

(d) $k : \mathbb{Z} \rightarrow \mathbb{Z}$, $k(n) = \frac{1}{2}n(n + 1)$;

(e) $s : \mathbb{Q} \rightarrow \mathbb{Z}$,

$$s(q) = \begin{cases} q & \text{if } q \in \mathbb{Z}, \\ 0 & \text{if } q \notin \mathbb{Z}; \end{cases}$$

(f) $t : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}$,

$$t(A) = \begin{cases} \pi & \text{if } A = \emptyset, \\ \text{the smallest element of } A & \text{if } A \neq \emptyset. \end{cases}$$

- A2. For each of the given functions f , determine whether f is injective, whether f is surjective, and whether f is bijective. [You might find it helpful to make a table and/or sketch a graph.] Justify your answers carefully. If f is a bijection, find the inverse function f^{-1} .

(a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = \frac{1}{2}(n + 1)(n + 2)$.

(b) $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(n) = \frac{1}{2}(n + 1)(n + 2)$.

(c) $f : [0, 1] \rightarrow [0, 1]$, $f(x) = 1 - x^2$.

Here $[0, 1]$ is the closed unit interval, $[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

- (d) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$.
- (e) $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}, f((x, y)) = x + y$.

A3. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. Prove the following results about the composite function $g \circ f: A \rightarrow C$:

- (a) if $g \circ f$ is injective then f is injective;
- (b) if $g \circ f$ is surjective then g is surjective;
- (c) if f and g are both injective then $g \circ f$ is injective;
- (d) if f and g are both surjective then $g \circ f$ is surjective.

SECTION B

These are more challenging questions. There may be questions on the January exam of similar difficulty to these.

B1. Give examples of functions $f_1, f_2, f_3, f_4: \mathbb{N} \rightarrow \mathbb{N}$ such that

- (a) f_1 is injective but not surjective;
- (b) f_2 is neither injective nor surjective;
- (c) f_3 is both injective and surjective (i.e., is bijective);
- (d) f_4 is surjective but not injective.

In each case you should justify your answer fully.

B2. Determine whether each of the following statements is true or false. If it is true, prove it. If it is false, show this by giving a counterexample. (In giving a counterexample, you will need to specify appropriate sets A, B, C .)

- (a) For functions $f: A \rightarrow B$ and $g: B \rightarrow C$, if $g \circ f$ is injective then g is injective;
- (b) For functions $f: A \rightarrow B$ and $g: B \rightarrow C$, if $g \circ f$ is surjective then f is surjective;
- (c) For a function $f: A \rightarrow A$, if $f \circ f = \text{id}_A$ then f is bijective;
- (d) For a function $f: A \rightarrow A$, if $f \circ f = f$ then $f = \text{id}_A$;
- (e) For a function $f: A \rightarrow A$, if $f \circ f = f$ then $f \circ f \circ f = f$.

- B3. Let A, B sets with $A \subseteq B$. The inclusion function $i : A \rightarrow B$ is defined by $i(a) = a$ for all $a \in A$. Show that i is always injective. When is i surjective?
- B4. Let A, B, C be sets with $A \subseteq B$, and let $f : B \rightarrow C$ be a function. We define a new function $f|_A : A \rightarrow C$ (called the restriction of f to A) by $f|_A(a) = f(a)$ for all $a \in A$. Show that if f is injective then $f|_A$ is also injective. Find an example where $f|_A$ is injective but f is not injective.
- B5. Find a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$, and find its inverse function $g : \mathbb{Z} \rightarrow \mathbb{N}$.
- B6. Show that, for any set S , there is an injective function $S \rightarrow \mathcal{P}(S)$.
- B7. Let A and B be *finite* sets, with $|A| = m$ and $|B| = n$. Prove that
- (a) if there is an injective function $f : A \rightarrow B$ then $m \leq n$;
 - (b) if there is surjective function $f : A \rightarrow B$ then $m \geq n$;
 - (c) if there is a bijective function $f : A \rightarrow B$ then $m = n$.
- Deduce that, for a finite set S , there can never be a surjective function $S \rightarrow \mathcal{P}(S)$.
- B8. Let S be a set (not necessarily finite), and let $f : S \rightarrow \mathcal{P}(S)$ be a function. Let $X = \{s \in S : s \notin f(s)\}$, and suppose that there is some $x \in S$ with $f(x) = X$. Prove that
- (a) if $x \in X$, then $x \notin X$;
 - (b) if $x \notin X$, then $x \in X$.

Deduce that there is no surjective function $S \rightarrow \mathcal{P}(S)$.